

Nitin Kumar Shukla

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Summary

Integrated M.Tech AI student at Vellore Institute of Technology with hands-on experience in the full Software Development Life Cycle (SDLC), system-level data processing, and C++-driven algorithmic problem solving. Proficient in Python, Java, and C++ with exposure to software component design, debugging, testing, and deployment. Experienced in building end-to-end software systems with a focus on performance optimization, data validation pipelines, and scalable architecture. Eager to contribute to cross-functional engineering teams delivering reliable, high-quality software products for health-tech applications.

Education

Vellore Institute of Technology 2022 – 2027
Integrated M.Tech in Artificial Intelligence | Deep Learning, Data Mining, Cloud Architecture **CGPA: 8.14**

Kendriya Vidyalaya 2022
Class XII **81%**

Kendriya Vidyalaya 2020
Class X **87.65%**

Experience

South Eastern Coalfields Limited (SECL) — *Data Analyst Intern* Oct 2024 – Dec 2024

- Designed and implemented an end-to-end medical bill scrutiny system in Python and C++, managing the complete SDLC from requirements through deployment.
- Developed robust REST APIs and OCR pipelines to extract, parse, and process structured data from large scanned document datasets.
- Engineered rule-based validation and medical entity recognition modules, applying system-level design to automate real-time reimbursement workflows.
- Built anomaly detection and risk analysis components to flag high-risk claims, reducing manual overhead through algorithmic decision logic.
- Conducted comprehensive system testing, debugging, and documentation to ensure maintainability and seamless deployment across subsystems.
- Collaborated with cross-functional stakeholders to validate system behavior, investigate issues, and iterate on software components throughout the lifecycle.

Projects

Stock Market Prediction System — Real-Time Data Processing

Architected an ML pipeline in Python (Scikit-learn, Pandas) using Random Forest and XGBoost for trend forecasting. Designed scalable software components for data ingestion, engineering, and model deployment. Applied C++-style algorithmic thinking to optimize real-time indicator processing and implemented automated data cleaning scripts to ensure accuracy under volatile data conditions.

Deepfake Detection Using AI & ML

Developed a deep learning system using CNN architectures (TensorFlow/Keras, OpenCV) for large-scale video dataset analysis and media authentication. Focused on system-level performance optimization to reduce model inference latency for potential real-time deployment. Conducted rigorous testing and debugging across multiple model iterations to improve detection robustness in digital security contexts.

Technical Skills

- Languages:** C++, Python, Java, SQL, JavaScript (Basic), REST APIs
- ML & Data:** Scikit-learn, TensorFlow, Keras, Pandas, NumPy, ML Pipelines
- Cloud & DevOps:** AWS (S3, EC2), Kubernetes, Git, GitHub, Real-time Data Processing
- Core Concepts:** SDLC, OS Fundamentals, System & Component Design, Debugging & Testing, Statistics, Cross-team Collaboration
- Engineering:** Performance Optimization, Algorithmic Problem Solving, Software Maintenance, Issue Investigation

Certifications

- Kubernetes for Absolute Beginners – Packt (2025)
- Applied Machine Learning in Python – University of Michigan (2023)
- Cloud Computing – NPTEL (2023)